Comparing Inference-Time Methods for Natural-Language Mathematical Proofs

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Abstract

We compare inference-time scaffolding methods for mathematical proof generation against plain sampling at matched cost. On 70 problems (60 IMO-ProofBench + 10 frontier-tier 2026 research items), six base models, four architectures, and three automated judges under a ~\$1,000 budget: (1) the generator-verifier-revisor pipeline does not beat pass@3 under either strict judge ($\Delta = -0.08$, problem-clustered 95% CI $[-0.25, +0.08]$, p=0.36); (2) seeded-ideator-plus-pipeline shows a positive effect concentrated at reasoning=max on cheap models (cluster $\Delta = +0.60$, p=0.005; partial coverage, no cell survives Holm correction); (3) judge choice flips conclusions: the lenient judge calls "pass" on 37% of strict-fail cases; (4) pass-rate degrades from 86% at pre-competition to 0% at research-frontier; (5) pooling 3,178 trials yields 32 strict-judge candidate passes on 2026 research problems, 23 on a single likely-contaminated item. The recommendation: draw more samples before adding pipeline depth, keep reasoning on, and audit with judges of different leniency.

1 Introduction

Language models have moved from saturating school-math benchmarks to plausibly contributing to research mathematics. AlphaProof achieved IMO silver in 2024 [Hubert et al., 2025]; gold-tier systems from DeepMind [Google DeepMind, 2025], OpenAI [Wei et al., 2025], and Harmonic [Achim et al., 2025] followed in 2025; and the Erdős repository now lists dozens of model-assisted contributions [Tao, 2026, Erdős Problems, n.d.].

Two methodological gaps motivate this work. First, comparing a multi-call pipeline against pass@1 is unfair: the proper baseline is pass@n matched to the pipeline's call budget, not pass@1. We use measured per-trial cost to set this matching (§3.1). Second, evaluation of free-form proofs at scale relies on automated judges, and the gap between lenient and strict judges is large enough to flip qualitative conclusions about which architecture wins.

A third constraint shaped every choice: a ~\$1,000 OpenRouter budget, which kept us off frontier-tier generators and judges and forced partial coverage of the most expensive architecture.

We do not propose a new method. We characterize a six-by-four-by-three space (model $\times$ architecture $\times$ judge) on a 70-problem benchmark and audit what holds up under paired bootstrap CIs and permutation tests. Contributions:

1. A token-cost-fair, statistically tested comparison of pass@k against generator-verifier-revisor pipelines and seeded-ideator variants, stratified by difficulty.
2. Quantification of judge-induced variance: cell-level CIs on the strict-vs-lenient false-positive rate.

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3. A frontier-solves audit pooling 3,178 trials with per-problem breakdowns.
4. Reproducible per-experiment artifacts totaling ∼3,200 trial rows.

2 Related work

Capability arc. The trajectory from GSM8K and MATH [Hendrycks et al., 2021] through IMO competitions includes AlphaProof [Hubert et al., 2025], Deep Think [Google DeepMind, 2025], Aristotle [Achim et al., 2025], and an open-source agentic framework [Huang and Yang, 2025]. Through early 2026, models produced solutions on the Erdős repository [Tao, 2026, Feng et al., 2026a], on FrontierMath open problems [Glazer et al., 2024], and via Aletheia [Feng et al., 2026b].

Benchmarks. IMO-ProofBench [Luong et al., 2025]: 30 Basic + 30 Advanced olympiad proofs with a hardened grader (r=0.96 to human judges). We augment with ten 2026 research problems (Erdős 333, 397, 654, 659, 1051; FirstProof 4, 5, 6, 10 [Abouzaid et al., 2026]; Epoch's Ramsey-hypergraph problem) and call the union PB+R26.

Architectures. Pass@k [Brown et al., 2020, Chen et al., 2021]: sample k solutions, take best. Self-consistency [Wang et al., 2022] and chain-of-thought [Wei et al., 2022] are established baselines. Generator-verifier-revisor pipelines appear in Aletheia [Feng et al., 2026b] and Huang–Yang [Huang and Yang, 2025]. We restrict to natural-language proofs; we do not study formal verification, evolutionary search [Novikov et al., 2025, Georgiev et al., 2025], or hybrid neuro-symbolic systems.

3 Methodology

3.1 Architectures

We implement four architectures in a single pipeline with shared model-call infrastructure, using prompts derived from Aletheia [Feng et al., 2026b] and Huang–Yang [Huang and Yang, 2025]. Final grading uses the IMO-ProofBench judge prompt, restricted to $\{0, 1, 6, 7\}$.

- **Pass@k.** Run k independent generations; take max judge score. We sweep $k \in \{1, 3\}$ in the main comparison, $k \in \{1, 3, 5, 7, 9\}$ in the scaling experiment. The `generate` mode produces 3 branches per trial (i.e., pass@3).
- **Generator-verifier-revisor (`full`).** Up to three verifier-revisor iterations; early-stop on `VERDICT: correct`.
- **Seeded ideator (`seed_generate`).** An ideation call produces three sketch directions; each is realized as a full solution.
- **Ideator + pipeline (`seed_full`).** Each ideated seed runs through the verifier-revisor loop.

Empirical cost matching. Median measured per-trial cost gives `full` at $0.78\times$ `generate`, `seed_generate` at $1.00\times$, and `seed_full` at $1.36\times$—far short of the nominal $3\times$ and $9\times$ (verifier/revisor calls are shorter and `full` early-stops on easy problems). Cost-matching: `full` and `seed_generate` $\approx$ pass@3; `seed_full` $\approx$ pass@4–5, not pass@9.

3.2 Models, reasoning, and dataset

Six cost-feasible base models: DeepSeek-v4-flash, -pro, Gemini-3-flash-preview, Gemma-4-31B-IT, GPT-OSS-120B, Qwen3.6-35B-A3B. Gemma-4 and GPT-OSS additionally run with reasoning=max. Table 1 shows the difficulty stratification.

3.3 Judges

Three judges score every architecture trial: DeepSeek-v4-flash (canonical), DeepSeek-v4-pro (strict audit), and Gemini-3-flash-preview (lenient audit). Section 4 reports the GradingBench calibration.

Table 1: Problem set and difficulty tiers. PB = ProofBench; R26 = 2026 research problems.

Tier	Source	n
pre-competition (d=0)	PB-Basic	30
competition-hard (d=1)	PB-Advanced + Erdős 397	31
research-easy (d=2)	Erdős 333, 654, 659; FirstProof 10	4
research-medium (d=3)	Erdős 1051; FirstProof 5	2
research-hard (d=4)	FirstProof 4, 6	2
research-frontier (d=5)	Ramsey hypergraphs	1

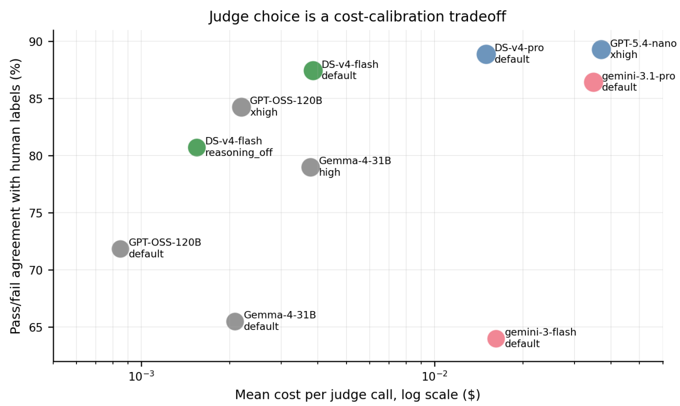

Figure 1: Judge cost–calibration Pareto front on GradingBench (n=200). DeepSeek-v4-flash (green, top-left cluster) is the only high-agreement judge below $0.01/call.

3.4 Statistical methodology

Paired comparisons use within-(model, reasoning, problem) score differences. *Problem-clustered* CIs (used for headlines) resample whole problems with replacement; 5,000 resamples, seed 20260507. p-values are sign-flip permutation tests (10,000 permutations); the cluster variant flips at the problem level. For §5.4's eight-test family we report Holm-corrected p-values. Pass-rate CIs use Wilson; zero-pass cells use the one-sided 95% Clopper–Pearson upper bound.

4 Calibration

Judge calibration on GradingBench. We tested ten judge configurations on 200 GradingBench problems (prompt restricted to $\{0, 1, 6, 7\}$). DeepSeek-v4-flash achieves 87.4% pass-agreement at $0.004/call, on the Pareto frontier of accuracy vs. cost (Figure 1). The next-best judges cost 3.8–9.5× more; Gemini-3-flash-preview agrees only 64% at $0.016/call. Full results are in Appendix A.

Generator selection on AnswerBench. Thirteen model–reasoning configurations on AnswerBench-50 (50 verifiable-answer problems). Reasoning is the dominant variable: GPT-OSS-120B jumps from 58% to 74%, Gemini-3-flash from 74% to 90%. Six models cleared 60% accuracy; all six enter the architecture sweep (Appendix B).

5 Main results

5.1 Architecture comparison

Table 2 shows pooled results across base models on PB+R26 under the v4-flash judge. Coverage is complete (6/6) for `generate`, `seed_generate`, `full`; partial (3/6) for `seed_full`.

The pass@1 → pass@3 step alone moves the mean +0.39 and pass-rate +5 pp—larger than every architecture-vs-architecture delta. Most apparent lift against pass@1 is the sample-size effect.

Table 2: Architecture comparison (pooled, v4-flash judge). Top: aggregate statistics with Wilson pass-rate CIs. Bottom: paired contrasts with problem-clustered bootstrap.

Mode	cost	coverage	n	mean	pass rate
pass@1	0.33×	6/6	613	1.97	0.28 [0.25, 0.32]
`generate` (= pass@3)	1.00×	6/6	543	2.31	0.33 [0.29, 0.37]
`seed_generate`	1.00×	6/6	555	2.29	0.33 [0.29, 0.37]
`full`	0.78×	6/6	550	2.24	0.32 [0.28, 0.36]
`seed_full` *(partial)*	1.36×	3/6	344	2.58	0.37 [0.32, 0.42]
Contrast		n / problems	Δ	cluster 95% CI	p
`full` − `generate`		533 / 70	−0.08	[−0.25, +0.08]	0.36
`seed_gen` − `generate`		539 / 70	−0.03	[−0.21, +0.16]	0.78
`seed_full` − `gen.` *(partial)*		337 / 70	+0.32	[+0.09, +0.55]	0.006

Table 3: Difficulty-stratified pass rates under the v4-flash judge. For zero-pass cells: one-sided 95% Clopper–Pearson upper bound.

Tier	n_{prob}	n_{trial}	mean	pass	95% interval
pre-competition	30	231	6.02	0.86	Wilson [0.81, 0.90]
competition-hard	31	1,507	2.04	0.29	Wilson [0.26, 0.31]
research-easy	4	115	1.50	0.23	Wilson [0.16, 0.31]
research-medium	2	57	0.14	0.02	Wilson [0.003, 0.09]
research-hard	2	55	0.00	0/55	upper $\leq$ 5.4%
research-frontier	1	27	0.00	0/27	upper $\leq$ 10.6%

The `full` − `generate` contrast (full coverage) is null (Δ=−0.08, p=0.36). The pooled `seed_full` − `generate` contrast is significant but rests on a non-representative subset and is concentrated in the reasoning=max stratum (§5.4).

5.2 Difficulty-stratified analysis

Pass-rate degrades sharply with difficulty (Figure 2; Table 3).

Architecture lift is null on hard tiers under strict judges. `full` − `generate` is null at every tier (R26-only Δ=+0.06, p=1.0). `seed_full` − `generate` on R26 is +0.48 [−0.03, +1.14] (n=42, cluster p=0.38), directionally positive but not significant. Figure 4 shows the lift-by-tier forest plot.

5.3 Judge choice flips the cheap-model comparison

The same raw outputs scored by Gemini-3-flash-preview tell a different story (Figure 3). Under Gemini, four of six models show +0.14 to +0.83 lift from the pipeline; the strict judges show no such pattern. On the 1,440 cells where all three judges scored: strict judges agree on 95.5% [94.4, 96.5] of pass/fail decisions. P(Gemini pass | v4-flash fail) = 0.37 [0.34, 0.40] over n=987 strict-fail cases—a systematic offset, not noise [Zheng et al., 2023]. The reverse, P(Gemini fail | v4-flash pass) = 0.035, over n=453. We use v4-flash for headline numbers.

Generator-judge family check. The v4-flash judge shares family with two generators (DeepSeek-v4-flash, -pro). For `seed_full` − `generate`, same-family Δ=+0.15 [−0.49, +0.79]; different-family Δ=+0.35 [+0.09, +0.62]. The lift is concentrated in cross-family generators, so judge-family bias does not drive the headline (Appendix J).

5.4 Where seeded-ideator-plus-pipeline helps

The pooled `seed_full` − `generate` lift (+0.32, p=0.006) is over a non-representative model subset and heavily confounded by reasoning. Stratifying (Table 4):

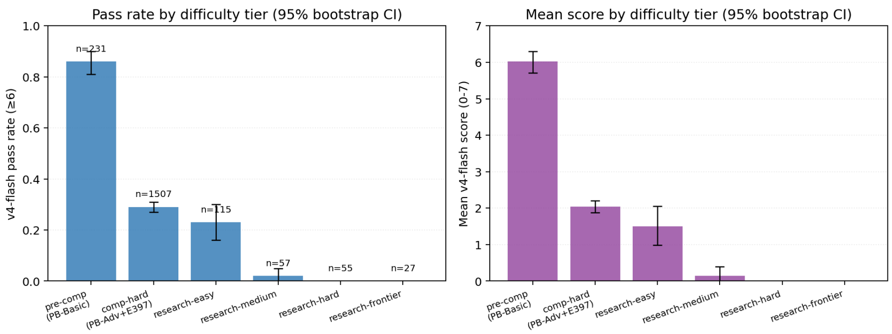

Figure 2: Pass rate (left) and mean score (right) by difficulty tier with 95% bootstrap CIs. Strict-judge pass rate falls from 86% at PB-Basic to 0% at research-hard and above.

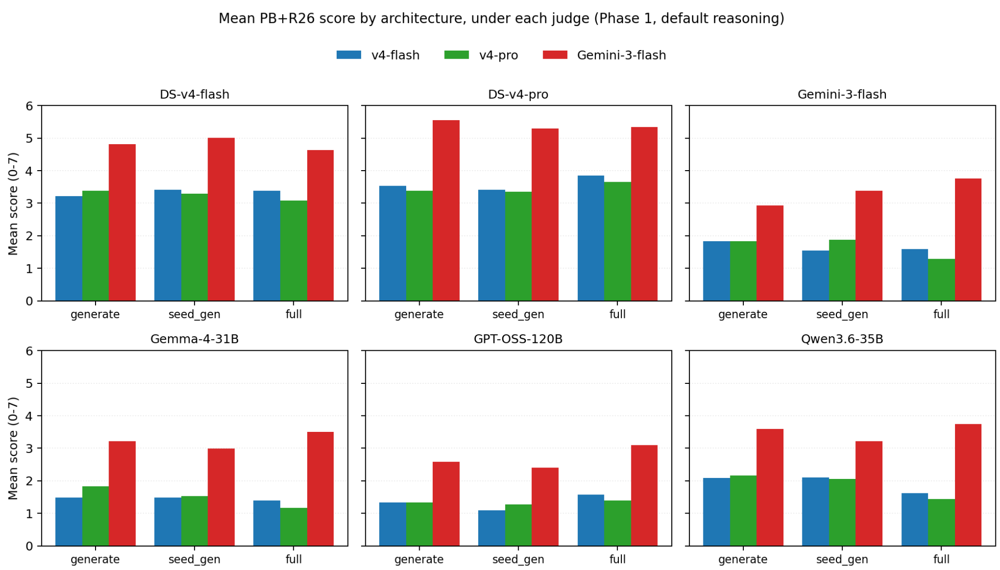

Figure 3: Mean PB+R26 score by architecture under each of three judges, for each base model (Phase 1, default reasoning). The lenient judge (Gemini, red) systematically inflates pipeline scores on weaker models.

The lift is entirely driven by reasoning=max cells. Eight per-cell paired tests (Appendix H) yield no cell surviving Holm correction at α=0.05 (smallest Holm-$p = 0.15$); the pooled-stratum cluster contrast survives (p=0.005).

Cost-matched comparison. At measured 1.36× generate cost, `seed_full` is closer to pass@4–5. Pass@5 means under v4-flash are 3.05 (Gemma max) and 2.83 (GPT-OSS max) vs. `seed_full` 3.31 and 3.00: a small edge for cheap models at reasoning=max. DeepSeek-v4-flash's pass@5 of 4.18 cleanly beats its `seed_full` of 3.49. Pure scaling wins for stronger generators.

5.5 Frontier candidate passes

We use "candidate pass" throughout: every count is an automated-judge pass not validated by an expert. Pooling all 3,178 v4-flash-judged trials across architecture (2,098) and scaling (1,080) experiments, Table 5 shows per-problem results.

Excluding FirstProof-10: 9 v4-flash candidate passes, 5 on Erdős-654 (spanning four base models), plus single cells on six other problems including the only strict candidate passes of Erdős-1051 and FirstProof-6 (both DeepSeek-v4-flash pass@7). For Erdős-1051, Gemini awards 14 passes against the strict judges' single (v4-flash) and zero (v4-pro)—a sharp illustration of judge fragility. The

Table 4: `seed_full` − `generate` lift stratified by reasoning mode (problem-clustered CIs).

Stratum	n/problems	Δ	cluster 95% CI	p
default reasoning	203 / 70	+0.13	[−0.14, +0.42]	0.39
reasoning=max	134 / 70	**+0.60**	**[+0.21, +1.03]**	**0.005**

Table 5: R26 candidate passes per problem, pooling all experiments. Each problem has ∼33 cells. †FirstProof-10 likely inflated by leaked solution direction.

Problem	Difficulty	cells	v4-flash		Gemini	
			pass	rate	pass	rate
FirstProof 10†	res.-easy	33	23	70%	8	24%
Erdős 654	res.-easy	33	5	15%	6	18%
Erdős 333	res.-easy	33	1	3%	2	6%
Erdős 659	res.-easy	33	1	3%	7	21%
Erdős 397	comp.-hard	33	1	3%	1	3%
FirstProof 5	res.-med.	33	1	3%	1	3%
Erdős 1051	res.-med.	33	1	3%	14	42%
FirstProof 6	res.-hard	33	1	3%	5	15%
FirstProof 4	res.-hard	32	0	0%	1	3%
Ramsey	res.-front.	33	0	0%	3	9%

defensible reading of any single strict-judge candidate pass at research-medium-or-harder difficulty is "candidate for expert review," not "confirmed solve."

6 Secondary results

Reasoning effects exceed architecture effects. Reasoning-on adds $+0.78$ to $+1.60$ to v4-flash mean score (Gemma and GPT-OSS; Appendix D), 3–5× larger than architecture deltas. This is well-established [Wei et al., 2022, OpenAI, 2024]; we present it as confirmation, not discovery.

Pass@k scales significantly through k=7. The scaling experiment computes pass@n by exhaustive enumeration over all $\binom{M}{n}$ subsets (Figure 5). At reasoning=max, pass@7 (∼3.1–3.2) matches `seed_full` (∼3.0–3.3) within 95% CIs. For DeepSeek-v4-flash at default reasoning, pass@7=4.54 is well above its `seed_full` mean of 3.49. Slope does not flatten by k=7.

Cross-model role-swap is null. Varying which model fills each pipeline role in `seed_full` at reasoning=max yields eight conditions spanning 2.96–3.63 mean; only GPT-OSS-as-verifier (3.63) beats both random baselines (3.19, 3.34). The "diversity helps" hypothesis is not supported (Appendix F).

Coverage asymmetry. Three base models (DeepSeek-v4-pro, Gemini-3-flash-preview, Qwen3.6) lack `seed_full` runs due to budget constraints. Closing this gap is the single largest follow-up the budget left on the table.

7 Limitations

Benchmark contamination. Most R26 solution-disclosure dates fall in 2026. Several generators have post-training cutoffs that overlap (Gemma-4: March 31; DeepSeek-v4: April 24). GPT-OSS-120B (released August 2025) is provably contamination-free for every R26 solution. Solve patterns for GPT-OSS and DeepSeek look broadly similar, which we read as weak evidence that contamination is not the dominant signal.

Judging. Calibrated to ∼87% pass-agreement on GradingBench (IMO-class items); judge accuracy on Erdős/First-Proof material is an extrapolation. Section 5.5's candidate-pass counts of 1/33 are candidates for expert review, not confirmed mathematics.

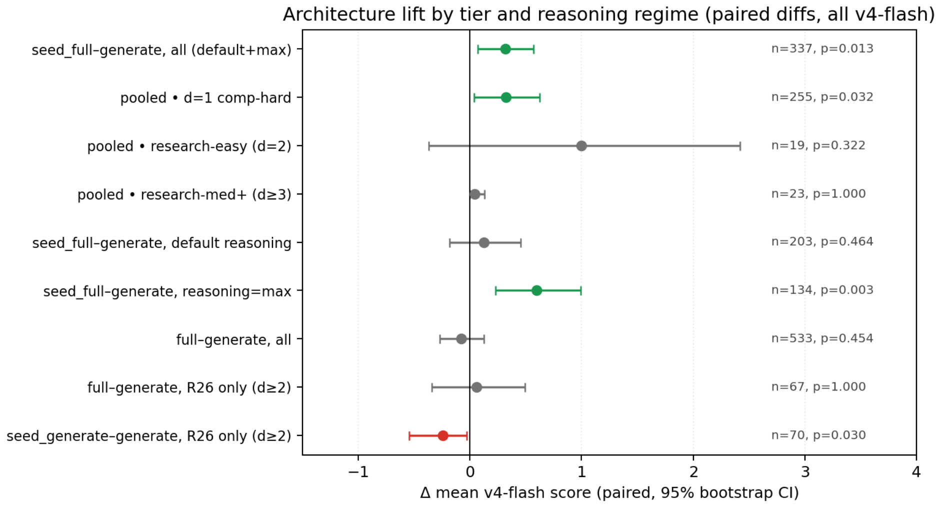

Figure 4: Architecture lift (paired Δ in v4-flash mean score) stratified by tier and reasoning regime. Green: significant at $p < 0.05$; grey: null; red: significant negative.

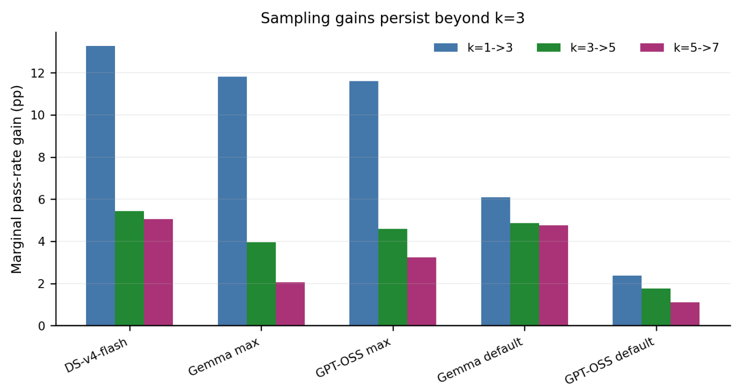

Figure 5: Marginal pass-rate gains from additional samples. The k=1→3 step dominates, but k=3→5 and k=5→7 persist, especially for DS-v4-flash and cheap models at default reasoning.

Generalization. Our models top out at DeepSeek-v4-pro. We do not test frontier-tier models on the full sweep, though a small probe (Appendix E) suggests Kimi-k2.6 deserves a future look.

Dataset size. n=70 is small; only ten problems are research-grade, with n=2–4 per tier at research-medium-and-harder.

Single-judge audit. The cleanest pipeline-lift result (§5.4) is v4-flash only; a v4-pro audit would strengthen the claim.

Multiple comparisons. Per-cell tests do not survive Holm at α=0.05; the pooled reasoning=max stratum contrast (p=0.005) is the load-bearing positive.

AI-assistance. Drafts were prepared with LLM assistance for prose and analysis-script scaffolding. All numerical results were computed by the referenced analysis scripts.

8 Discussion

Three takeaways.

The verifier-revisor pipeline does not beat call-matched sampling. Across 533 paired cells, `full` − `generate` is null ($\Delta{=}{-}0.08$, $p{=}0.36$). Seeded-ideator-plus-pipeline shows a positive effect concentrated in reasoning=max cheap-model cells ($\Delta{=}{+}0.60$, $p{=}0.005$); at default reasoning the effect is null; per-cell tests do not survive Holm correction. The recommendation: draw more samples before adding pipeline depth, keep reasoning on, use a calibrated judge.

Audit with judges of different leniency. Strict judges agree on 95.5% of decisions; Gemini calls "pass" on 37% of strict-fail cases. A two-judge audit is a cheap discipline.

Frontier candidate passes are concentrated and judge-fragile. Pooling 3,178 trials yields 32 strict-judge R26 candidate passes; 23 on FirstProof-10 (likely contaminated). Excluding it: 9 candidate passes, 5 on Erdős-654. A candidate pass from a cheap model on a research-medium-or-harder problem is a flag for expert review, not a theorem.

Budget and follow-ups. Results came from ~\$800 of inference. ~\$20 of v4-pro grades on the four reasoning=max `seed_full` cells would harden §5.4's result. A Phase-1 `seed_full` extension on three missing base models closes the largest coverage gap. A fair pass@k-vs-pipeline comparison at frontier models would test whether the cheap-model finding generalizes.

Resources and audit. Architecture sweep: 2,098 trials × 6 base models. Scaling: 1,080 trials. Aggregate API spend ~\$800 against a \$1,000 budget. Known data-quality issue: original Phase 1 seeded-ideator runs had ∼10–28% malformed-ideator outputs that silently fell back to pass@3, which would *understate* any seeded-ideator effect; we patched the bug for Phase 2 onward.

References

Mohammed Abouzaid, Andrew J. Blumberg, Martin Hairer, et al. First proof. *arXiv preprint arXiv:2602.05192*, 2026. URL `https://1stproof.org/`.

Tudor Achim, Alex Best, Alberto Bietti, et al. Aristotle: IMO-level automated theorem proving. *arXiv preprint arXiv:2510.01346*, 2025.

Tom B. Brown, Benjamin Mann, Nick Ryder, et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33:1877–1901, 2020.

Mark Chen, Jerry Tworek, Heewoo Jun, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.

Erdős Problems. Erdos problems database. `https://www.erdosproblems.com/`, n.d.

Tony Feng, Trieu Trinh, Garrett Bingham, et al. Semi-autonomous mathematics discovery with Gemini: A case study on the Erdős problems. *arXiv preprint arXiv:2601.22401*, 2026a.

Tony Feng, Trieu H. Trinh, Garrett Bingham, et al. Towards autonomous mathematics research (Aletheia). *arXiv preprint arXiv:2602.10177*, 2026b.

Bogdan Georgiev, Javier Gómez-Serrano, Terence Tao, and Adam Zsolt Wagner. Mathematical exploration and discovery at scale. *arXiv preprint arXiv:2511.02864*, 2025.

Elliot Glazer, Ege Erdil, Tamay Besiroglu, et al. FrontierMath: A benchmark for evaluating advanced mathematical reasoning in AI. *arXiv preprint arXiv:2411.04872*, 2024.

Google DeepMind. Advanced version of Gemini with Deep Think officially achieves gold-medal standard at the International Mathematical Olympiad. Google DeepMind blog, July 21, 2025, 2025.

Dan Hendrycks, Collin Burns, Saurav Kadavath, et al. Measuring mathematical problem solving with the MATH dataset. *arXiv preprint arXiv:2103.03874*, 2021.

Yuntian Huang and Lin F. Yang. Winning gold at IMO 2025 with a model-agnostic verification-and-refinement pipeline. *arXiv preprint arXiv:2507.15855*, 2025.

Thomas Hubert et al. Olympiad-level formal mathematical reasoning with reinforcement learning (AlphaProof). *Nature*, 651:607–613, 2025. doi: 10.1038/s41586-025-09833-y.

Thang Luong, Dongmin Hwang, Huy Hoang Nguyen, et al. Towards robust mathematical reasoning (IMO-Bench). *arXiv preprint arXiv:2511.01846*, 2025. URL `https://imobench.github.io/`.

Alexander Novikov, Nghi Vũ, Marvin Eisenberger, et al. AlphaEvolve: A coding agent for scientific and algorithmic discovery. *arXiv preprint arXiv:2506.13131*, 2025.

OpenAI. Learning to reason with LLMs (o1). OpenAI blog, September 12, 2024, 2024.

Terence Tao. Primitive sets and von Mangoldt chains: Erdős problem #1196 and beyond. What's New blog, May 3, 2026, 2026.

Xuezhi Wang, Jason Wei, Dale Schuurmans, et al. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.

Alexander Wei, Sully Hsu, and Noam Brown. OpenAI's experimental reasoning LLM achieves gold-medal performance on IMO 2025. Announcement thread, July 19, 2025, 2025.

Jason Wei, Xuezhi Wang, Dale Schuurmans, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35, 2022.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *Advances in Neural Information Processing Systems*, 36, 2023.

A GradingBench full table

Ten judge configurations on n=200 from GradingBench, sorted by pass-agreement at 6.

Judge	Reasoning	n	pass$\geq$6	recall	spec	F1	r	$/call	lat.
GPT-5.4-nano	xhigh	196	89.3%	70%	98%	0.800	0.711	$0.037	152s
DS-v4-pro	default	198	88.9%	83%	92%	0.825	0.756	$0.015	299s
DS-v4-flash	default	199	87.4%	79%	91%	0.800	0.756	$0.004	81s
Gemini-3.1-pro	default	199	86.4%	95%	82%	0.816	0.872	$0.035	22s
GPT-OSS-120B	xhigh	178	84.3%	90%	82%	0.788	0.770	$0.002	62s
DS-v4-flash	reas. off	192	80.7%	56%	92%	0.654	0.591	$0.002	13s
Gemma-4-31B	high	200	79.0%	92%	73%	0.734	0.776	$0.004	126s
GPT-OSS-120B	default	199	71.9%	74%	71%	0.622	0.513	$0.001	9s
Gemma-4-31B	default	200	65.5%	98%	50%	0.642	0.629	$0.002	19s
Gemini-3-flash	default	200	64.0%	98%	48%	0.633	0.606	$0.016	4s

B AnswerBench-50 generator calibration

Model	Config	Accuracy	Notes
DeepSeek-v4-pro	default	94% (47/50)	top accuracy
GPT-5.4-nano	xhigh	92% (45/49)	1 hung
Gemini-3-flash	xhigh (synth)	90% (45/50)	
DeepSeek-v4-flash	default	88% (44/50)	best value
Qwen3.6-35B-A3B	default	80% (40/50)	
GPT-OSS-120B	xhigh	74% (37/50)	+16pp vs default
Gemini-3-flash	default	74% (37/50)	
Gemma-4-31B-IT	default	68% (34/50)	reas. off
GPT-OSS-120B	default	58% (29/50)	
DS-v4-flash	reasoning off	50% (25/50)	−38pp

Model × Mode	Reas.	v4-flash	v4-pro	Gemini
DS-v4-flash × generate	def	3.22	3.39	4.81
DS-v4-flash × seed_gen	def	3.41	3.30	5.01
DS-v4-flash × full	def	3.38	3.09	4.63
DS-v4-flash × seed_full	def	3.49	3.19	5.83
DS-v4-pro × generate	def	3.53	3.38	5.56
DS-v4-pro × seed_gen	def	3.41	3.36	5.30
DS-v4-pro × full	def	3.85	3.66	5.35
Gemini-3-flash × generate	def	1.83	1.83	2.93
Gemini-3-flash × seed_gen	def	1.54	1.87	3.39
Gemini-3-flash × full	def	1.59	1.29	3.76
Gemma-4 × generate	def	1.49	1.83	3.21
Gemma-4 × seed_gen	def	1.49	1.53	2.99
Gemma-4 × full	def	1.39	1.17	3.51
Gemma-4 × seed_full	def	1.71	1.67	4.26
Gemma-4 × generate	max	2.74	–	–
Gemma-4 × seed_full	max	3.31	–	–
Gemma-4 × full	max	2.17	–	–
GPT-OSS × generate	def	1.34	1.33	2.59
GPT-OSS × seed_gen	def	1.09	1.27	2.40
GPT-OSS × full	def	1.58	1.39	3.10
GPT-OSS × seed_full	def	1.41	1.37	3.99
GPT-OSS × generate	max	2.30	–	–
GPT-OSS × seed_full	max	3.00	–	–
GPT-OSS × full	max	2.43	–	–
Qwen3.6 × generate	def	2.09	2.17	3.60
Qwen3.6 × seed_gen	def	2.10	2.06	3.21
Qwen3.6 × full	def	1.62	1.44	3.74

Model	Mode	default	max	Δ	95% CI
Gemma-4	generate	1.49	2.74	+1.25	[+0.74, +1.77]
Gemma-4	seed_full	1.71	3.31	+1.60	[+1.07, +2.14]
GPT-OSS	generate	1.34	2.30	+0.96	[+0.46, +1.46]
GPT-OSS	seed_full	1.41	3.00	+1.59	[+1.05, +2.13]

C Per-cell means (architecture × judge × model)

D Reasoning effect on PB+R26

E Expensive-models probe

n=12 problems, default reasoning.

Model	Accuracy	$/run	Notes
Kimi-k2.6	100% (10/10)	$0.097	2/12 hung
Qwen3.6-Max	92% (11/12)	$0.172	dominated
Gemini-3.1-Pro	83% (10/12)	$0.250	dominated
GPT-5.4	50% (6/12)	$0.054	artifact at default

Condition	n	mean	pass rate
random_run1	69	3.19	0.45
random_run2	70	3.34	0.49
x_ideate_gemma	68	3.09	0.44
x_ideate_oss	70	2.96	0.41
x_revise_gemma	69	3.48	0.49
x_revise_oss	70	3.41	0.49
x_verify_gemma	69	3.00	0.43
x_verify_oss	70	**3.63**	**0.51**

Pair	Agreement	95% CI
v4-flash vs v4-pro	95.5%	[94.4, 96.5]
v4-flash vs Gemini	73.7%	[71.5, 76.0]
v4-pro vs Gemini	72.9%	[70.6, 75.3]

F Roleswap detail

G Judge agreement rates

H Per-cell Holm-corrected tests

I Empirical cost per architecture

J Generator-judge family stratification

Same family = DeepSeek generators; different family = all others. v4-flash judge throughout.

Conditional flip	Estimate	95% CI	n
P(Gemini pass | v4-flash fail)	0.368	[0.338, 0.398]	987
P(Gemini fail | v4-flash pass)	0.035	[0.020, 0.053]	453

Cell	n	Δ	p_{raw}	p_{Holm}
Gemma max, seed_full − gen	70	+0.57	0.039	0.275
GPT-OSS max, seed_full − gen	64	+0.63	0.046	0.278
Gemma max, full − gen	70	−0.57	0.019	0.152
GPT-OSS max, full − gen	65	+0.17	0.70	1.00
DS-v4-flash def, seed_full − gen	66	+0.15	0.71	1.00
DS-v4-pro def, full − gen	60	+0.23	0.57	1.00
Gemma def, seed_full − gen	69	+0.14	0.56	1.00
GPT-OSS def, seed_full − gen	68	+0.09	0.86	1.00

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Mode	n	mean $	median $	p90 $	ratio
generate (= pass@3)	558	0.084	0.040	0.224	1.00×
seed_generate	560	0.084	0.041	0.226	1.00×
full	560	0.066	0.030	0.129	0.78×
seed_full	349	0.114	0.071	0.215	1.36×

Contrast	Same family Δ [CI]	Different family Δ [CI]
full − gen	+0.11 [−0.29, +0.54]	−0.14 [−0.36, +0.09]
seed_gen − gen	−0.05 [−0.47, +0.38]	−0.02 [−0.22, +0.17]
seed_full − gen	+0.15 [−0.49, +0.79]	+0.35 [+0.09, +0.62]

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